Explainable Federated Learning for Telecom Churn

Capstone prototype — simulated clients; not a production privacy guarantee.

This interface launches the experiment script. Run the experiment once before the mentor demo so the outputs are already available. The first run may need internet access to download the dataset.

Run experiment

```
Loading local dataset: data/Telco-Customer-Churn.csv

Dataset shape: (7043, 21)
Target distribution (0=No churn, 1=Churn):
Churn
No     0.735
Yes    0.265
Name: proportion, dtype: float64

Training centralized Logistic Regression baseline...

CENTRALIZED RESULTS
  accuracy: 0.7381
 precision: 0.5043
    recall: 0.7834
        f1: 0.6136
   roc_auc: 0.8413

Centralized classification report:
              precision    recall  f1-score   support

    No churn       0.90      0.72      0.80      1035
       Churn       0.50      0.78      0.61       374

    accuracy                           0.74      1409
   macro avg       0.70      0.75      0.71      1409
weighted avg       0.80      0.74      0.75      1409


Training federated logistic regression with FedAvg...

Federated simulation: 3 clients
Round 01/12 | global training log-loss: 0.6112
Round 03/12 | global training log-loss: 0.5423
Round 06/12 | global training log-loss: 0.4987
Round 12/12 | global training log-loss: 0.4603

MODEL COMPARISON (actual results from this run)
                              model  accuracy  precision  recall     f1  roc_auc
       Centralized Logistic Regression    0.7381     0.5043  0.7834 0.6136   0.8413
Federated Logistic Regression (FedAvg)    0.7786     0.7422  0.2540 0.3785   0.8363

Calculating permutation feature importance for centralized model...

Calculating SHAP explanations for centralized model...
SHAP global and individual explanation files generated.

Example prediction:
 actual_churn  centralized_churn_probability  centralized_prediction  federated_churn_probability  federated_prediction
            0                      0.115472                       0                     0.121078                     0

Saved outputs:
 - outputs/model_comparison.csv
 - outputs/confusion_matrices.png
 - outputs/feature_importance.csv
 - outputs/feature_importance.png
 - outputs/shap_global_importance.csv
 - outputs/shap_global_importance.png
 - outputs/shap_individual_explanation.csv
 - outputs/shap_individual_explanation.png
 - outputs/example_prediction.csv

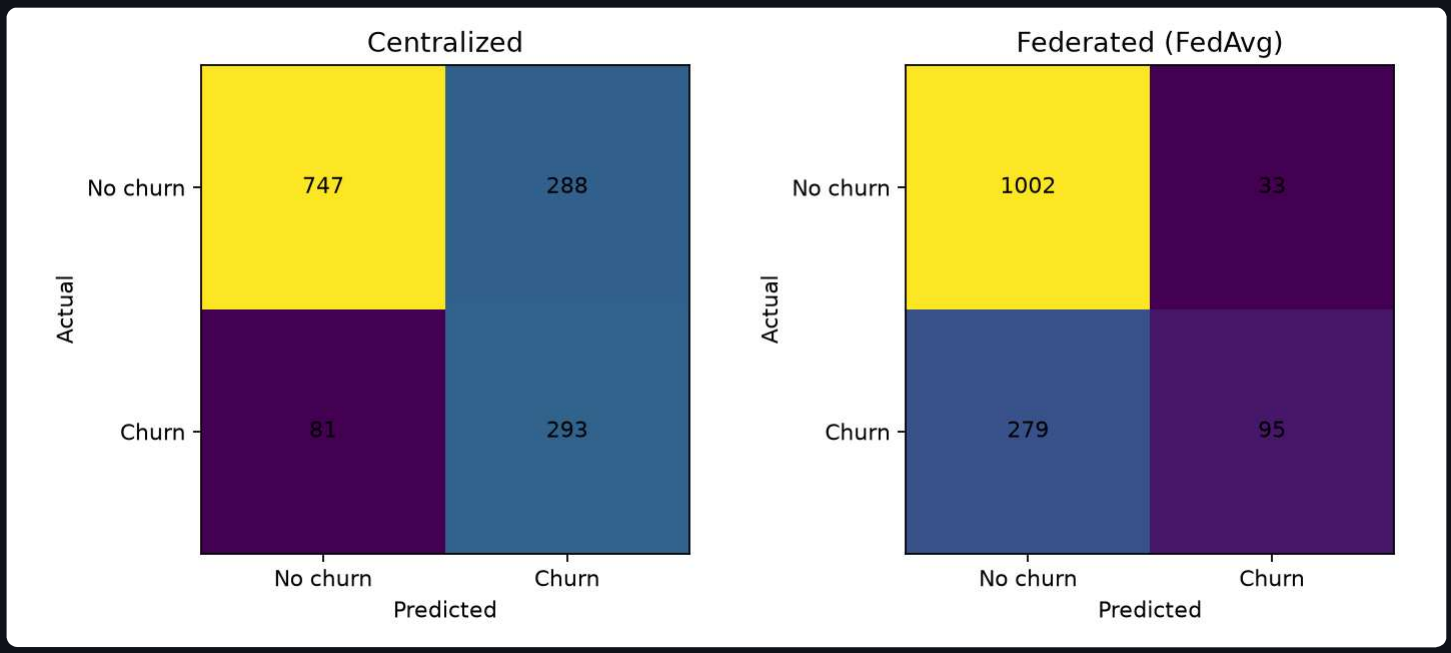
Note: the preprocessing transformer was fitted before the simulated client split.
This is a prototype limitation; report it transparently in your capstone.
```

Experiment completed.

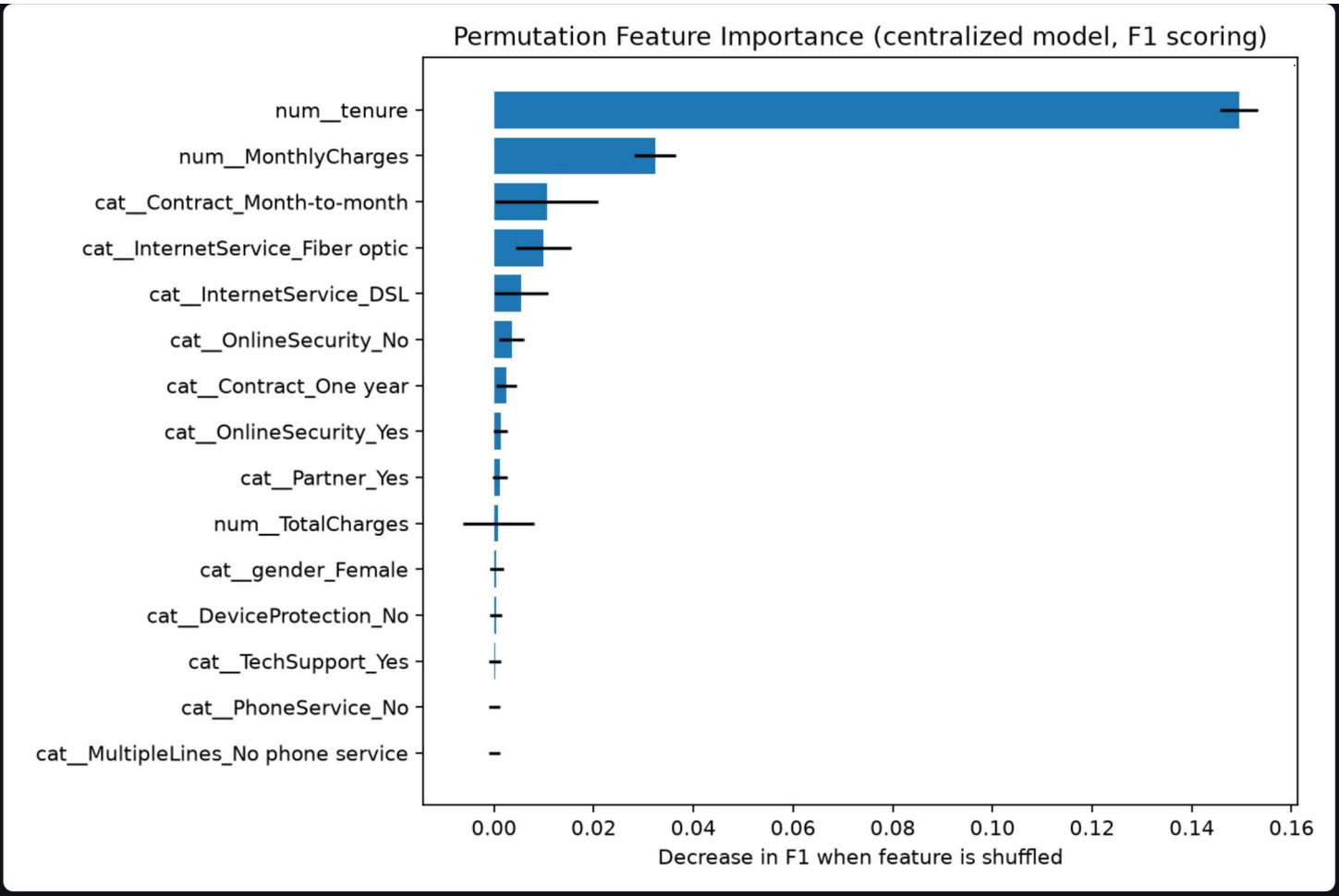
Centralized vs Federated results

	model	accuracy	precision	recall	f1	roc_auc
0	Centralized Logistic Regression	0.7381	0.5043	0.7834	0.6136	0.8413
1	Federated Logistic Regression (FedAvg)	0.7786	0.7422	0.254	0.3785	0.8363

Confusion matrices



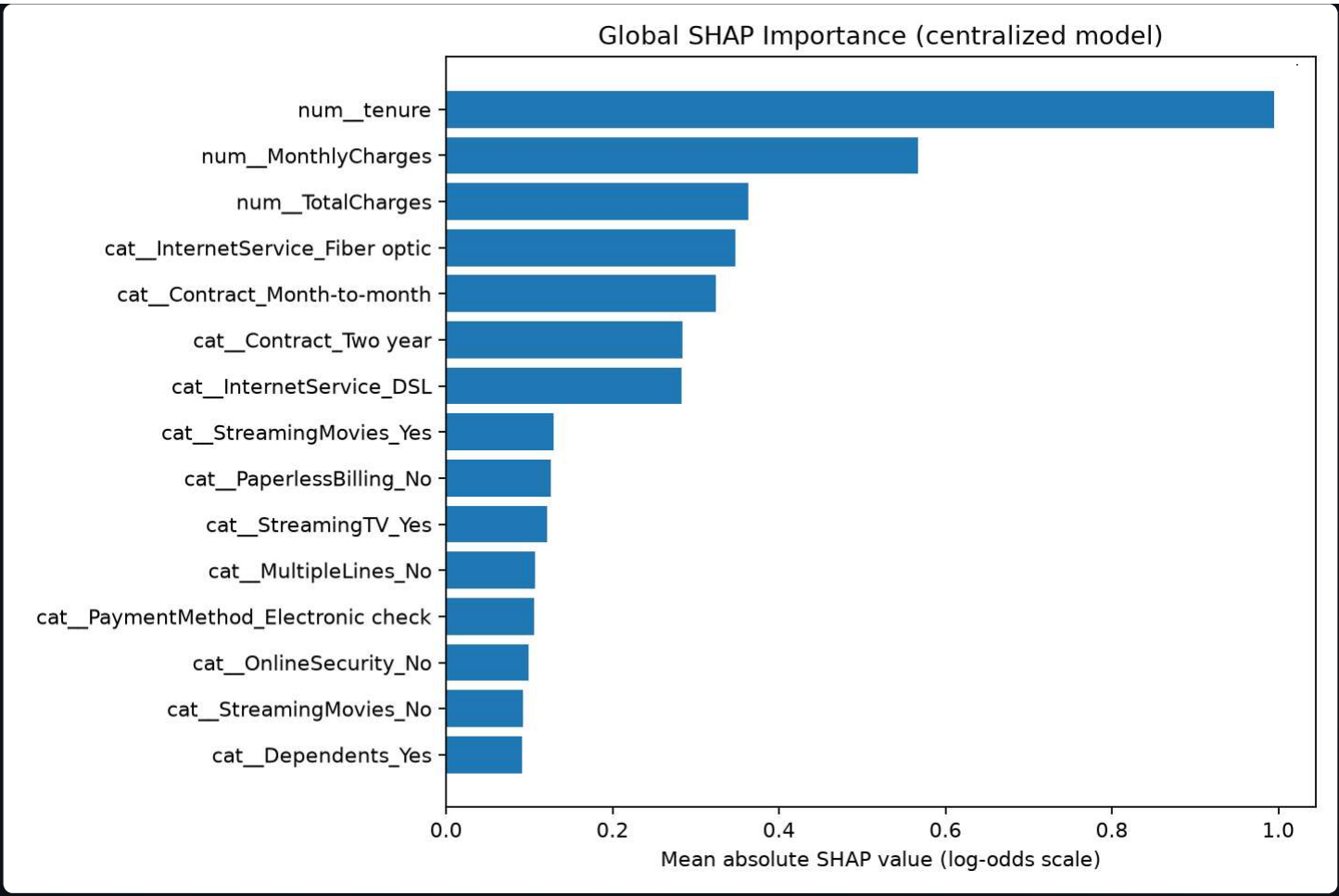
Explainability: permutation feature importance



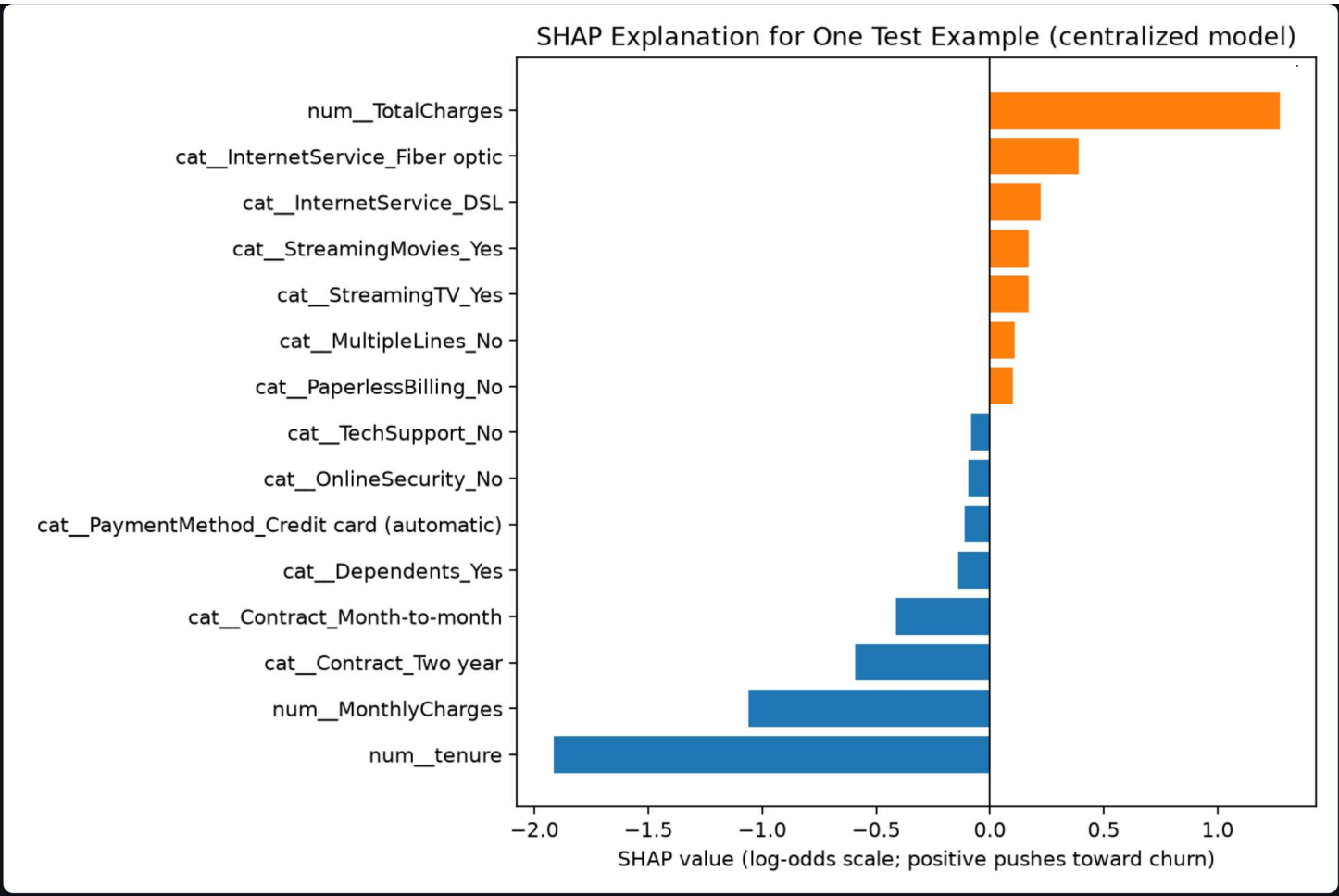
Example prediction

	actual_churn	centralized_churn_probability	centralized_prediction	federated_churn_probability	federated_prediction
0	0	0.1155	0	0.1211	0

Global SHAP Importance



Individual SHAP Explanation



Global SHAP Values

	feature	mean_absolute_shap_value
10	cat__MultipleLines_No	0.1067
11	cat__PaymentMethod_Electronic check	0.1053
12	cat__OnlineSecurity_No	0.0989
13	cat__StreamingMovies_No	0.0927
14	cat__Dependents_Yes	0.0908
15	cat__StreamingTV_No internet service	0.0899
16	cat__InternetService_No	0.0899
17	cat__OnlineSecurity_No internet service	0.0899
18	cat__TechSupport_No internet service	0.0899
19	cat__DeviceProtection_No internet service	0.0899